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Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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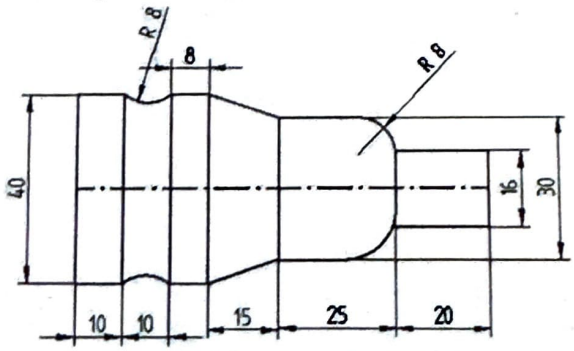
Academic year 2024-2025 (EVEN Sem)

DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	1st April 2025	Maximum Marks	10 + 50
Course Code	IM343AI	Duration	120 Min
Sem	IV Semester	CIE –I	
CAD/CAM & ROBOTICS			

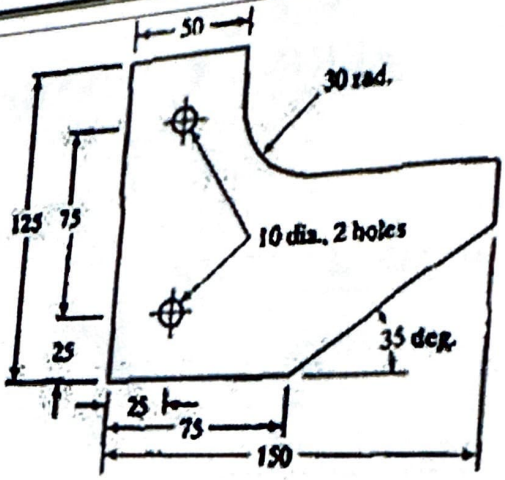
Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	Part – A			
1.1	List any two advantage of using CAD over manual drafting?	02	L2	CO1
1.2	How does DNC differ from CNC? State any two.	02	L1	CO2
1.3	List any two reasons for implementing CAD in engineering and product design?	02	L2	CO2
1.4	What are the key components of a DNC system?	02	L2	CO1
1.5	What are two essential elements of an NC system?	02	L1	CO2
	Part –B			
1	Discuss the various applications of computers in the design process with its various stages and their importance in engineering design.	10	L1	CO2
2	According to the type of machine, NC machine are classified into three types explain the classification with the help of Case examples.	10	L2	CO1
3	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. Using G & M Codes 	10	L4	CO2
4	Develop a complete APT Part Program for machining the given part. (Profile Milling only). Tool-Slot drill Cutter of ϕ10mm, Cutting Speed-1500, Feed- 60, Depth of cut=6mm	20	L4	CO1

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Academic year 2024-2025 (EVEN Sem)



BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Ma x Marks	04	06	-	-	04	06	-	-	-	--
	Test		20	30	--	-	10	10	-	30	-	--

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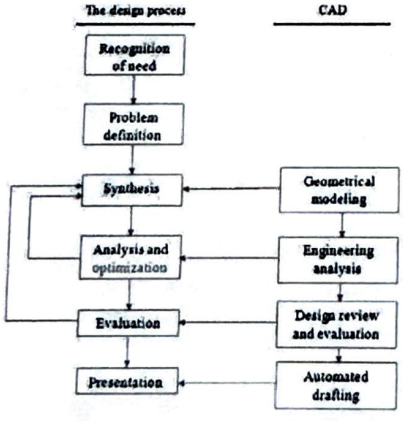
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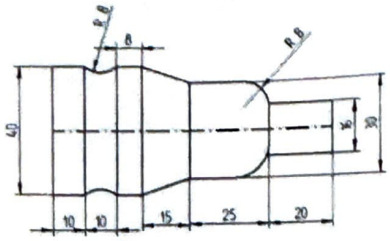
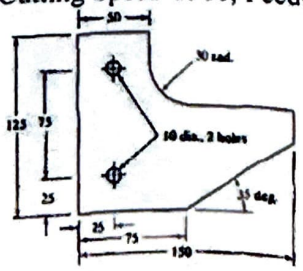
Academic year 2024-2025 (EVEN Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	1st April 2025	Maximum Marks	10 + 50
Course Code	IM343AI	Duration	120 Min
Sem	IV Semester	CIE –I	
CAD/CAM & ROBOTICS			

Note:
1. Answer all the Questions.

Sl. No.	Questions	M
	Part – A	
1.1	• CAD allows precise measurements and error-free drawings. • Designs can be edited or updated quickly without redrawing the entire layout.	02
1.2	• CNC (Computer Numerical Control) controls a single machine, while DNC (Direct Numerical Control) can control multiple machines from a central computer. • In CNC, programs are loaded directly onto the machine; in DNC, the central computer sends instructions directly to machines in real-time.	02
1.3	• Speeds up the design process and reduces development time. • Offers 2D and 3D views, helping in better understanding and communication of designs.	02
1.4	• Central Computer: Manages and distributes part programs to multiple machines. • Communication Lines/Networks: Connect the central computer to CNC machines for data transmission.	02
1.5	• Program of Instructions: The coded data that defines operations (often in G-code). • Machine Control Unit (MCU): Interprets the instructions and controls the machine's movements.	02
	Part –B	
1	Discuss the various applications of computers in the design process with its various stages and their importance in engineering design.  There are four steps in the CAD design process: • Geometric modeling (in place of synthetic step): This is a mathematical technique that is done by using simulation software. It shows the object from the math formula through computer graphics. • Engineering analysis (in place of analysis and optimization): This is the process of checking the strength of moving objects to determine how they will work in the real world.	10

	• Design review and evaluation (replaces evaluation): Design is reviewed and evaluated with the help of CAD software. Also, the feasibility of the design is checked. • Automatic drafting (replaces presentation): The design is checked to see if it meets product specifications. The design is then saved automatically.	10
2	The classification of NC machine tool systems can be done in three ways: 1. According to the type of machine: Point-to-point, Straight cut and contouring (continuous path) 2. According to the programming method: incremental versus absolute 3. According to the type of control loops: Open-loop versus closed-loop	10
3	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. Using G & M Codes  G21 G98 G28 U0 W0 M06 T01 M03 S1000 G00 X30 Z05 G71 U1 R0.5 G71 P10 Q20 U0 W0 F80 N10 G01 X16 F20 G01 Z-20 G03 X30 Z-28 R8 G01 X30 Z-45 G01 X40 Z-60 G01 X40 Z-68 N20 G03 X40 Z-78 R8 G01 X42 G28 U0 W0 M05 M30	10
4	Develop a complete APT Part Program for machining the given part. (Profile Milling only). Tool-Slot drill Cutter of ϕ10mm, Cutting Speed-1500, Feed- 60, Depth of cut=6mm  Post processor-MACHIN/MILL,1	20

Spindle speed – 1500rpm
Feed rate – 60mm/min
Tolerance – 0.01mm/min
Thickness- 15mm
Cutter dia – 20mm
Cutter location - -50.0, -50.0, 50.0

PART NO/1	SPINDL/1500
MACHIN/MILL,01	FEDRAT/60
INTOL/0.01	COOLANT/ON
OUTOL/0.01	FROM/P0
CUTTER/20.0	GO/TO, L1 TO PL1, TO P5
SPINDL/1000	GOLFT/L1,PAST, P2
FEDRAT/75	GORGT/P2, PAST, L2
COOLNT/ON	GOFWD/L2, PAST, P3
STRT=POINT/-50.0, -50.0, 50.0	GODWN/L3, PAST, L4
P0=POINT/0.0,0.0, 0.0	GOFWD/C1, PAST, P5
P1=POINT/0,0, -6	GORGT/L4, PAST, P6
P2=POINT/0,125, -6	GORGT/L5,PAST, P7
P3=POINT/50,125, -6	GORGT/L6, PAST, P8
P4=POINT/50,100, -6	GORGT/L7, PAST, P1
P5=POINT/75,75, -6	GOTO/P0
P6=POINT/150,75, -6	SPINDL/OFF
P7=POINT/150,52.5, -6	COOLNT/OFF
P8= POINT/75,0, -6	FINI
P9=POINT/80,100, -6	
L1=LINE/P1, P2	
L2=LINE/P2,P3	
L3=LINE/P3, P4	
L4=LINE/P5, P6	
L5=LINE/P6, P7	
L6=LINE/P7, P8	
L5=LINE/P8, P1	
C1=CIRCLE/CENTER, P9, RADIUS, 30	
PL1=PLANE/P1, P2, P3	

Academic year 2023-2024 (EVEN Sem)

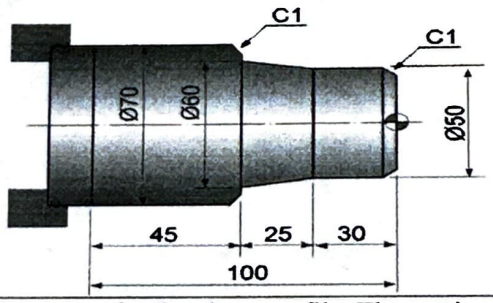
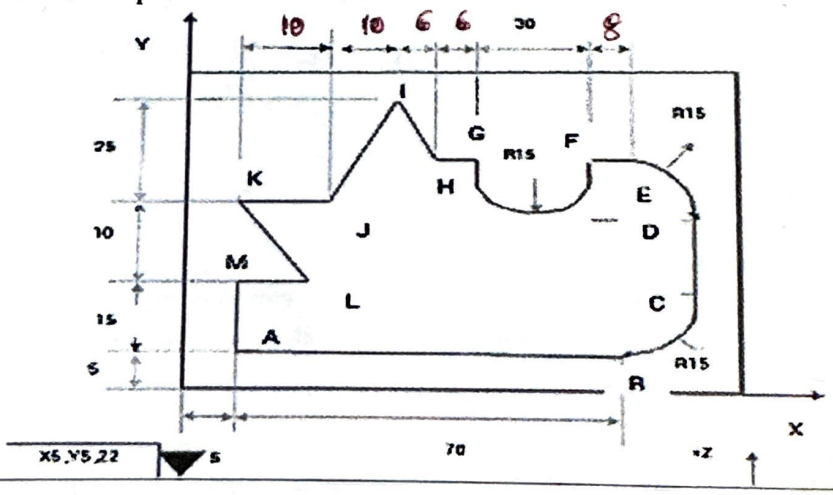
DEPARTMENT OF

INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**18th June 2024**	**Maximum Marks**	**50**
Course Code	**IM343AI**	**Duration**	**90 Min**
Sem	**IV Semester**	**CIE – I**	
CAD/CAM & ROBOTICS			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1	When is it appropriate to use the NC Program in manufacturing environment? Elaborate.	10	L1	CO1
2	State are the differences between the following? a) absolute and incremental coordinate system b) machine zero and program zero	10	L2	CO1
3	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. C1=3X45^0 	15	L3	CO1
4	Write a CNC Milling program for the given profile. The cutting tool used is slot drill cutter and the depth of cut is 5mm. 	15	L3	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	50	--	--	--	10	10	30	--	--	--

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RV Educational Institutions ®
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Academic year 2022-2023 (Even Sem)

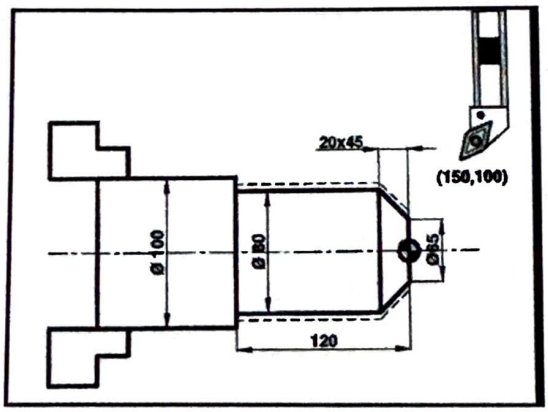
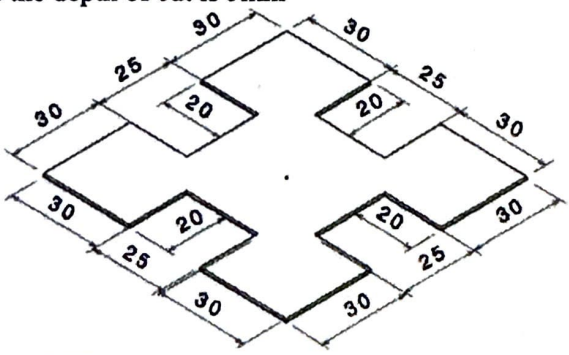
DEPARTMENT OF

INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	**6th July 2023**	**Maximum Marks**	**50**
Course Code	**21IM43**	**Duration**	**90 Min**
Sem	**IV Semester**	**CIE - I**	
CAD/CAM & ROBOTICS			

Note:

1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
1.	State the differences between Lathe, Numeric control machine and Computer numeric control with respect to quality, time, cost and automation?	10	L1	CO1
2.	When is it appropriate to use the NC Program in manufacturing environment? Elaborate.	10	L2	CO1
3.	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. 	15	L4	CO2
4.	Write a CNC milling program for the given profile. The cutting tool used is slot drill cutter and the depth of cut is 5mm 	15	L4	CO2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	20	30	--	--	10	10	--	30	--	--